

DSAA3073: Theories in Data Science

Week 3 Open Investigation

How Large Is the Policy Search We Must Protect?

From a clinical threshold rule to finite-sample behavior and VC structure

Basic analysis. A pre-registered clinical triage policy

Use only the learning-framework, ERM, finite-class, and VC tools in the Week 3 Learning Sheet

A hospital uses a scalar risk score $x \in [0, 100]$ to decide whether a patient should receive extended observation after discharge. The label $y = 1$ means that a serious complication occurred within seven days. Before examining the audit data, the clinical committee registers 32 threshold rules

$$h_{t(x)} = \mathbf{1}_{\{x \geq t\}}, \quad t \in \mathcal{T}, \quad |\mathcal{T}| = 32.$$

The zero-one loss records disagreement with the observed outcome. An i.i.d. audit sample $S \sim \mathcal{D}^{800}$ is drawn from the intended patient population. ERM selects $\hat{h}_S$ with empirical risk 0.125. The hospital requests a distribution-free statement at confidence 0.95.

Basic Task 1. Declare the learning problem and the selected object

Specify $\mathcal{X}, \mathcal{Y}, \mathcal{H}, \ell, \mathcal{D}$ and S . Define population and empirical risk, write the ERM rule, and distinguish the observed number 0.125 from the unobserved population risk. Explain why applying a fixed-hypothesis Hoeffding statement only after ERM has selected a threshold would leave a logical gap.

Basic analysis. Protect the data-dependent threshold

Take the union before selection

Basic Task 2. Construct and interpret the finite-class certificate

Use the Week 3 finite-class simultaneous-deviation theorem with $m = 800$, $|\mathcal{H}| = 32$, and $\delta = 0.05$. Compute the uniform radius α , an upper bound for $R(\hat{h}_S)$, and the corresponding ERM excess-risk guarantee relative to the best population rule in the registered class. State the clinical conclusion conditionally and explain why it does not prove that threshold triage is the best possible policy outside $\mathcal{H}$.

Thinking investigation. Formalize the class-guarantee relation

The number of parameter values and the number of finite-sample behaviors are different objects

Represent the investigation by

$$(\mathcal{X}, \mathcal{Y}, \mathcal{H}, \ell, \mathcal{D}, S, A, \mathcal{G}),$$

where A is ERM and $\mathcal{G}$ is the statistical guarantee. The basic model assumes a finite, pre-registered class. The next investigations change the policy class or the realizability assumption while keeping the risk notation fixed.

Investigation 1. Replace the finite threshold grid by all real thresholds

Let $\mathcal{H}_{\text{thr}} = \{h_t : t \in \mathbb{R}\}$. Decide whether the finite-cardinality union theorem can still be applied as written. For m distinct ordered scores, derive the exact number of restriction vectors, determine the VC dimension, and compare the result with the Sauer-Shelah bound. State precisely what this combinatorial expression does and does not yet prove statistically.

Thinking investigation. Enlarge the geometry of the policy

Two cut points create a different exact growth law

Investigation 2. Replace one threshold by one observation interval

Suppose extended observation is recommended only for an intermediate risk band: $h_{a,b}(x) = \mathbf{1}_{\{a \leq x \leq b\}}$ with $a \leq b$. Determine the VC dimension by giving the lower-bound construction and universal obstruction. Then derive the exact growth function on m ordered scores and compare its order with that of thresholds. Under realizability, state the resulting standard sample-complexity scale without inventing the hidden absolute constant.

Thinking investigation. Remove realizability

A perfect training fit no longer supports the faster bad-consistent argument

Investigation 3. Allow irreducible label noise in the registered finite class

Return to the 32 registered thresholds but suppose the same score can produce both labels with positive probability. Decide which Week 3 theorem fails and which theorem remains valid. Derive the surviving analytic ERM guarantee at general (m, K, δ) and compare its leading accuracy dependence with the consistent-realizable sample bound.

Synthesis. Return to the clinical decision

Explain how the guarantee changes as the policy class moves from a pre-registered finite grid, to all thresholds, to all intervals, and then to noisy labels. Recommend one defensible next step before clinical deployment.